

Ritesh Raj

📍 Sydney, Australia ✉ rajritesh03@gmail.com ☎ 0415745143 🔗 in/raj-ritesh 🌐 ritesh-raj-portfolio.vercel.app/

SUMMARY

AI Engineer specializing in Generative AI, NLP tools, and harness engineering. Expert in leveraging Python, Next.js, and NestJS to build virtual file systems, sandboxes, and autonomous loops that make agentic systems predictable, traceable, and stateful at scale.

EXPERIENCE

Senior Associate

PwC

May 2022 - Present

- Developed **Pathfinder**, an AI-powered regulatory compliance platform that reduced manual review effort by 60%.
- Built LLM-based pipelines in Python to extract metadata, references, definitions, and regulatory obligations from complex documents.
- Co-designed graph representations of regulations to map relationships between chapters, sections, and legal obligations.
- Contributed to AI-driven workflows that surfaced relevant compliance insights and actionable recommendations.
- Integrated pipelines into a full-stack architecture using Next.js, NestJS, GraphQL, and PostgreSQL.
- Developed **Information Document Analyzer (IDA)**, an AI-driven document processing solution reducing manual data-entry effort by 70%.
- Built extraction and field-inference capabilities using Azure Form Recognizer to automate data capture.
- Contributed to asynchronous processing workflows using RabbitMQ to ensure reliable task execution.
- Integrated extracted data with GraphQL APIs and PostgreSQL for secure storage and retrieval.
- Built React.js frontend components allowing users to easily review and manage document processing results.

Technology Analyst

Infosys

August 2015 - May 2022

- Developed Customer Consent Management features for **Open Banking** to enable secure data sharing with accredited third parties, improving transparency and compliance for 100K+ users. Built reusable React.js components, designed REST APIs with Node.js and Swagger, implemented JWT-based authentication, and utilized Redis for caching and Splunk for monitoring.
- Developed a self-service **Infrastructure Provisioning Platform** (HPaaS VPC) that reduced manual effort and costs by 60%, enabling developers to seamlessly configure cloud resources. Created dynamic tables and forms with React.js, integrated Azure AD authentication with a Node.js backend, and automated tasks using Bash scripting.
- Built a **Test Environment Management System (TEMS)** to accelerate release cycles by 20–25% and boost productivity by 30–40% by identifying configuration conflicts early. Developed dynamic forms and D3.js visualizations using React.js, integrated Jira APIs for real-time visibility, and implemented backend logic with Node.js and SQL Server.

CERTIFICATIONS

Microsoft Certified: Azure Administrator Associate

2019

SKILLS

AI / LLM: NLP, Vercel AI SDK, RAG, Agentic Pipeline (tools + orchestration), Prompt Engineering, Harness

Backend & Full Stack: React.js, Next.js, NestJS, Streamlit, Python

Data & Infrastructure: Docker, Kubernetes, PostgreSQL

Engineering: Design Patterns, Data Structures & Algorithms

EDUCATION

Bachelor in Computer Engineering

University of Mumbai • 2015